# 被街道重塑的大脑

## *The Brain That Learned London*



熟记伦敦街道，会怎样改变大脑？

**READ** 纯净阅读

**CHECK** 中文核对

**LEARN** 词汇记忆

**RECALL** 回想巩固

# 英文原文

Name two addresses and a London cabbie will rattle off the quickest route. That prowess is hard-won. To earn a licence, aspiring drivers must conquer "the Knowledge", an arduous examination on about 25,000 streets within six miles of Charing Cross. Hopefuls typically spend three to four years committing this labyrinth to memory, and only about half qualify.

A study published in PNAS in April 2000 scanned 16 male cabbies and 50 other men. The hippocampus is a seahorse-shaped structure vital to memory and navigation. In the cabbies, its rear, or posterior, portion was larger than in the other men, and its front, or anterior, portion smaller. Longer service went with a larger rear portion.

Yet the link left the cause uncertain. Perhaps an innate flair for spatial tasks lured people to the occupation, or perhaps stressful driving was the culprit. A December 2006 study dispelled the second notion: cabbies had more rear grey matter, the tissue dense with nerve cells, than bus drivers, confined to predetermined routes, of equivalent experience and stress.

A December 2011 longitudinal study tackled the first, scanning 79 trainees and 31 non-trainees at enrolment and three to four years later. Their brains began alike. Only the 39 who qualified exhibited an expansion of rear grey matter, so the sequence suggests learning preceded the enlargement.

The gain appeared to entail a trade-off: qualified drivers trailed the non-trainees on one test of recalling an intricate figure. The cohorts were small and exclusively male, yet the studies imply the healthy adult brain can remodel itself through intensive learning. Every street learned, it seems, occupied room once claimed by something else.

# 中文译文

只要说出两个地址，伦敦出租车司机就能迅速报出最快的路线。这份本领来之不易。为了取得执照，准司机必须攻克一项叫作“知识大考”的艰难考试，熟记距查令十字六英里以内约2.5万条街道。备考者通常花三到四年把这座街道迷宫牢记于心，最终只有约一半能够合格。

2000年4月发表于《美国国家科学院院刊》的一项研究，扫描了16名男性出租车司机和50名其他男性的大脑。海马体形似海马，对记忆和认路至关重要。与其他男性相比，出租车司机海马体的后部更大，前部则更小。从业时间越长，后部也越大。

然而，这种关联并不能确定原因。也许是天生擅长空间任务的人更容易被这个职业吸引，也许是紧张的驾驶工作造成了差异。2006年12月的一项研究排除了后一种解释：与驾驶经验、压力水平相当的公交司机相比，出租车司机海马体后部的灰质仍然更多。灰质是一种密布神经细胞的组织，而公交司机的行驶路线是预先固定的。

2011年12月的一项纵向研究开始检验前一种解释。研究者在79名学员报名时，以及三到四年后，分别扫描了他们和31名未参训者的大脑。各组起初的大脑状况相似，只有最终通过考试的39名学员，出现了海马体后部灰质增加。这一先后顺序提示，学习发生在灰质增大之前。

这种收获似乎也伴随着某种取舍：在一项回忆复杂图形的测试中，通过考试的司机表现不如未参训者。研究的样本群体较小，而且全部为男性，但结果仍提示，健康的成人大脑可以通过高强度学习重塑自身。看来，每记住一条街道，都像是占用了原先属于其他事物的一点空间。

# 核心词汇

先记这22项。连着下方的原文短语一起回想。

**prowess** n.
高超本领
*That prowess is hard-won.*

**arduous** adj.
艰难费力的
*an arduous examination*

**commit** v.
此处指把……牢记于心
*committing this labyrinth to memory*

**qualify** v.
取得资格；合格
*and only about half qualify.*

**hippocampus** n.
海马体
*The hippocampus*

**navigation** n.
辨路；导航
*vital to memory and navigation.*

**posterior** adj.
后部的
*its rear, or posterior, portion*

**anterior** adj.
前部的
*its front, or anterior, portion*

**uncertain** adj.
无法确定的
*Yet the link left the cause uncertain.*

**innate** adj.
天生的
*an innate flair for spatial tasks*

**spatial** adj.
空间的
*spatial tasks*

**dispel** v.
消除；打消
*study dispelled the second notion:*

**predetermined** adj.
预先确定的
*confined to predetermined routes*

**equivalent** adj.
相当的；等同的
*of equivalent experience and stress.*

**longitudinal** adj.
长期追踪同一群体的
*longitudinal study*

**expansion** n.
扩大；增加
*an expansion of rear grey matter*

**precede** v.
发生在……之前
*learning preceded the enlargement.*

**entail** v.
意味着；必然涉及
*to entail a trade-off:*

**trade-off** n.
取舍；有得有失
*appeared to entail a trade-off*

**cohort** n.
接受研究的一组人
*The cohorts were small*

**imply** v.
提示；意味着
*yet the studies imply*

**remodel** v.
重塑；重新构建
*can remodel itself*

# 文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

**rattle off** phr. v.
快速流利地说出

**hard-won** adj.
来之不易的

**licence** n.
执照；许可证

**aspiring** adj.
有志于成为……的

**conquer** v.
攻克；征服

**examination** n.
考试

**hopeful** n.
有望成功者；此处指备考者

**typically** adv.
通常

**labyrinth** n.
迷宫

**scan** v.
扫描检查

**vital** adj.
至关重要的

**portion** n.
部分

**flair** n.
天赋；才能

**lure** v.
吸引；诱使

**occupation** n.
职业

**stressful** adj.
造成精神压力的

**culprit** n.
问题的起因；罪魁祸首

**notion** n.
看法；想法

**dense** adj.
密集的

**nerve** n.
神经

**confine** v.
限制；使局限于

**tackle** v.
着手处理；检验

**trainee** n.
受训学员

**enrolment** n.
报名；入学

**exhibit** v.
表现出；显示

**sequence** n.
先后顺序

**enlargement** n.
增大

**trail** v.
落后于

**intricate** adj.
复杂精细的

**exclusively** adv.
全部；仅仅

**intensive** adj.
高强度的；集中的

**occupy** v.
占据

**claim** v.
此处指占用

# 回想与回顾

先遮住下方答案，再把四个词放回原文。

**01** To earn a licence, aspiring drivers must conquer "the Knowledge", an _____ examination on about 25,000 streets within six miles of Charing Cross.

**02** The _____ is a seahorse-shaped structure vital to memory and navigation.

**03** Yet the link left the cause _____.

**04** The gain appeared to entail a _____: qualified drivers trailed the non-trainees on one test of recalling an intricate figure.

## 文章主线

- 训练 伦敦出租车司机通常花三四年记住约2.5万条街道，只有约一半学员通过考试。
- 差异 海马体参与记忆与认路；司机的海马体后部比对照男性大、前部小，最初无法判断原因。
- 排查 与经验、压力相近但走固定路线的公交司机相比，出租车司机的海马体后部灰质仍更多。
- 追踪 学员与对照者起初脑部相似；三四年后，仅考过者海马体后部灰质增加，支持成人大脑可随学习改变。
- 边界 样本小且全为男性；考过者在一项图形记忆测试中比未参训者差，提示认路能力增强可能伴随记忆取舍。

参考答案　1 arduous 2 hippocampus 3 uncertain 4 trade-off